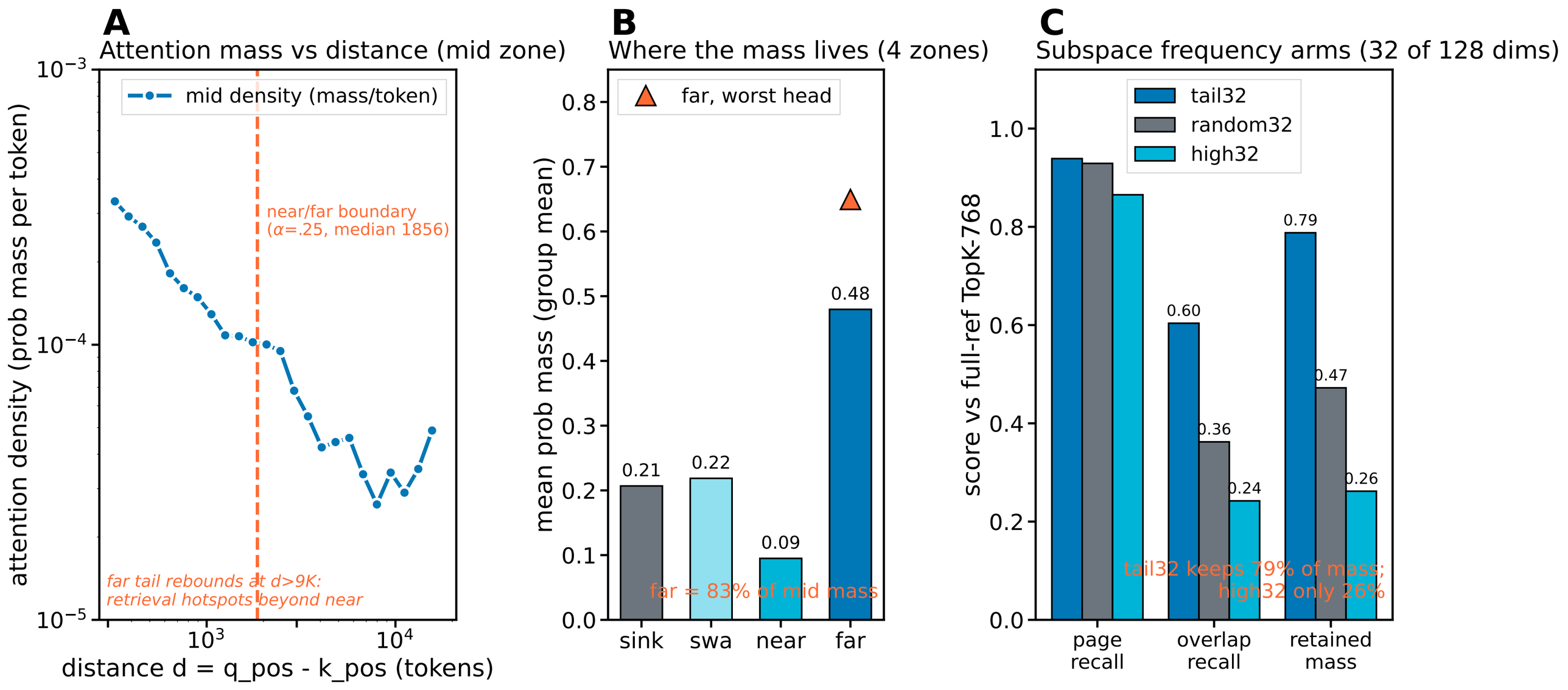
A
Attention mass vs distance (mid zone)
mid density (mass/token)
near/far boundary
(α=.25, median 1856)
far tail rebounds at d>9K:
retrieval hotspots beyond near
attention density (prob mass per token)
distance d = q_pos - k_pos (tokens)
B
Where the mass lives (4 zones)
far, worst head
mean prob mass (group mean)
0.21
0.22
0.09
0.48
far = 83% of mid mass
sink
swa
near
far
C
Subspace frequency arms (32 of 128 dims)
tail32
random32
high32
score vs full-ref TopK-768
0.60
0.36
0.24
0.79
0.47
0.26
tail32 keeps 79% of mass;
high32 only 26%
page recall
overlap recall
retained mass